

Stable rank based expressivity and generalization analysis of Low-Rank-Adapters

Nikita Okhotnikov

Scientific supervisor: Andrey Grabovoy

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Introduction

LoRA – most popular yet still understudied method of parameter-efficient fine-tuning

Problem

Existing theoretical estimates rely on algebraic rank and misalign with practical results

Observation

Empirical results show that it is not the rank that limits the performance of the adapters or lead to overfitting

Proposal

Derive the more practical bounds based on stable rank

Preliminary theory

Interpretable SR-Constrained Approximation Error Lower Bound

For matrix A stable rank is $sr(A) = \frac{\|A\|_F^2}{\|A\|_{op}^2} = \frac{\sum_i \sigma_i^2(A)}{\sigma_1^2(A)}$

Lemma (single matrix approximation lower bound)

A – target matrix, $\hat{A}$ – an estimate. $S_A = sr(A)$, $S_{\hat{A}} = sr(\hat{A})$, $S_{\hat{A}} \leq S_A$ then:

$$\|A - \hat{A}\|_F \geq \sigma_1(A) \left(\frac{\sqrt{S_A} - \sqrt{S_{\hat{A}}}}{1 + \sqrt{S_{\hat{A}}}} \right)$$

Lemma 2 (Optimal SR-constrained approximation)

Let $A_R = \sum_{i=1}^R \sigma_i u_i v_i^T$ be the optimal rank- R approx. of A , $S \in [1, R]$ the SR limit. $\mathcal{M}_{R,S}$ = matrices with rank $\leq R$, $sr \leq S$.

If $sr(A_R) \leq S$: $\hat{A} = A_R$. If $sr(A_R) > S$, the optimal $\hat{A} \in \mathcal{M}_{R,S}$ has:

$$\hat{\sigma}_1 = \frac{\sigma_1 + \sqrt{S-1} \sqrt{\sum_{j=2}^R \sigma_j^2}}{S}, \quad \hat{\sigma}_i = \gamma \sigma_i \quad (i \geq 2), \quad \gamma = \frac{\sqrt{S-1} \hat{\sigma}_1}{\sqrt{\sum_{j=2}^R \sigma_j^2}}$$

Minimal error: $\|A - \hat{A}\|_F^2 = \sum_{i=R+1}^D \sigma_i^2 + \Delta_S$,

$$\Delta_S = \frac{1}{S} \left(\sqrt{\sum_{i=2}^R \sigma_i^2} - \sqrt{S-1} \sigma_1 \right)^2 \quad (\text{If } sr(A_R) \leq S, \Delta_S = 0)$$

Fully connected network approximation

Notation

target FNN $\bar{f} := \text{FNN}_{\bar{L},D}(\cdot; (\bar{W}_l)_{l=1}^{\bar{L}}, (\bar{b}_l)_{l=1}^{\bar{L}})$,

frozen FNN $f_0 := \text{FNN}_{L,D}(\cdot; (W_l)_{l=1}^L, (b_l)_{l=1}^L)$,

adapted FNN $f := \text{FNN}_{L,D}(\cdot; (W_l + \Delta W_l)_{l=1}^L, (\hat{b}_l)_{l=1}^L)$

$b_l, \bar{b}_l, \hat{b}_l \in \mathbb{R}^D$, $W_l, \bar{W}_l, \Delta W_l \in \mathbb{R}^{D \times D}$, but $\Delta W_l = A_l B_l$, where $A_l \in \mathbb{R}^{D \times R}$ and $B_l \in \mathbb{R}^{R \times D}$ and $R \leq D$ – the LoRA rank. Additionally, we assume $\bar{L} = L$.

Theorem (FNN error bound)

Let $x \in \mathbb{R}^d$ be the random input with covariance matrix $\Sigma = \mathbb{E}[xx^T]$. Let $E_l = \bar{W}_l - W_l$ be the weight discrepancy. Let $E_{l,R}$ be its optimal rank- R approx. with singular values $\sigma_{l,1} \geq \dots \geq \sigma_{l,R}$. Assume $\hat{b}_l = \bar{b}_l$. Then there exist adapters ΔW_l with $\text{sr}(\Delta W_l) \leq S \ \forall l \in \overline{1, \bar{L}}$ such that the expected error is bounded by:

$$\mathbb{E} \|f(x) - \bar{f}(x)\|_2 \leq \beta \sum_{l=1}^L \left(\prod_{k=l+1}^L \|\bar{W}_k\|_2 \right) \mathcal{T}_l, \text{ where}$$

$$\beta = \max_{i \in [L]} \left(\left(\prod_{j=1}^i \|\bar{W}_j\| \right) \sqrt{\text{tr}(\Sigma)} + \sum_{j=1}^i \left(\prod_{k=j+1}^i \|\bar{W}_k\| \right) \|\bar{b}_j\| \right)$$

$$\text{and } \mathcal{T}_l = \sqrt{\sum_{j=R+1}^D \sigma_{l,j}^2 + \Delta_{S,l}}, \text{ where}$$

$$\Delta_{S,l} = \frac{1}{S} \left(\sqrt{\sum_{j=2}^R \sigma_{l,j}^2} - \sqrt{S-1} \sigma_{l,1} \right)^2 \text{ if } \text{sr}(E_{l,R}) > S, \quad 0 \text{ else}$$

Transformer Network Approximation

Notation

$TN_{L,D}$: L blocks, H -head self-attention + 2-layer FFN, no dropout/norm.
Target $\bar{f}$, frozen f_0 , adapted f (LoRA on all Q, K, V, O , both FFN layers, output).

$\mathcal{T}(G) = \sqrt{\sum_{i=R+1}^D \sigma_i^2(G) + \Delta_S(G)}$ – SR error from **Lemma 2**.

Theorem (Transformer error bound)

There exists a $\leq S$ SR adapter configuration such that

$$\mathbb{E} \|f(x) - \bar{f}(x)\|_F \leq \beta \sum_{l=1}^L \left(\prod_{k=l+1}^L \gamma_k \right) \mathcal{D}_l$$

- γ_k : Lipschitz constant of target block k w.r.t. input (via FFN + MHA)
- $\mathcal{D}_l = \mathcal{T}_{2,l} + \|\overline{W_{2,l}}\| \mathcal{T}_{1,l} + \|\overline{W_{2,l}}\| \|\overline{W_{1,l}}\| \sum_h \mathcal{D}_{att,l}^h$ (FFN + attention errors)
- $\mathcal{D}_{att,l}^h$: per-head distortion $\propto \mathcal{T}_O, \mathcal{T}_V, \mathcal{T}_Q, \mathcal{T}_K$ amplified by score sensitivity C_{score}

Upper-bound for LoRA Generalization Gap

Notation

Depth- T network, 1-Lipschitz $\sigma(\cdot)$, $\sigma(0) = 0$, LoRA on each layer $\Delta W_t = B_t A_t^T$.
 $\|x_i\|_2 \leq B_x$, $\|\Delta W_t\|_{op} \leq \Lambda_t$, $\text{sr}(\Delta W_t) \leq S_t$, $d_{\max} = \max_t(d_t)$.

Margin: $m(f_W(x), y) = f_W(x)_y - \max_{j \neq y} f_W(x)_j$

$$L_\gamma(W) = \mathbb{P}_{(x,y)}[m(f_W(x), y) \leq \gamma], \quad \hat{L}_\gamma(W) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[m(f_W(x_i), y_i) \leq \gamma]$$

Theorem (PAC-Bayesian generalization bound)

$\forall \gamma > 0$ and any $W = W_0 + \Delta W$ that satisfies the aforementioned conditions with probability at least $1 - \delta$ by the choice of training sample of size N :

$$L_0(W) \leq \hat{L}_\gamma(W) + \mathcal{O} \left(\sqrt{\frac{B_x^2 T^2 (d_{\max} + \ln(TN)) \left(\prod_{t=1}^T \bar{\beta}_t^2 \right) \sum_{t=1}^T \frac{\Lambda_t^2 S_t}{\bar{\beta}_t^2} + \log(N/\delta)}{\gamma^2 N}} \right)$$

Scales as $\mathcal{O}(\sqrt{S/N})$ unlike previously known best $\mathcal{O}(\sqrt{r/N})$.

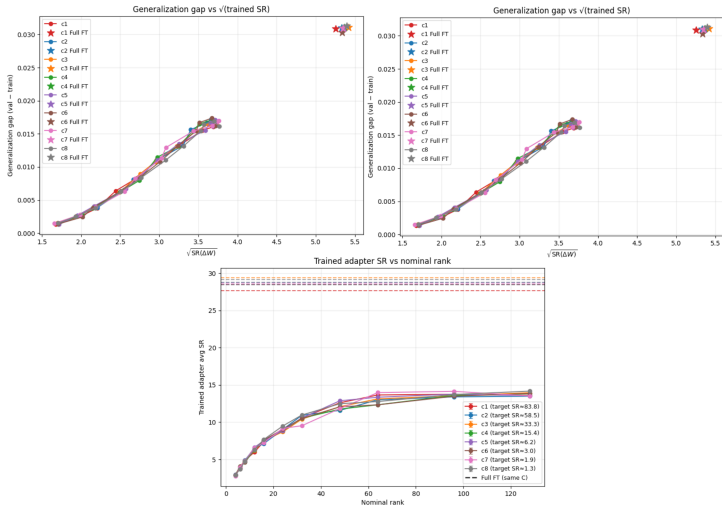
Experiments

Controlled small scale setting

Initialize 10-layer MLP $\bar{f}_0$ (teacher), copy it's weights to f_0 (student)

Shift $\bar{f}_0 \rightarrow \bar{f}$ by adding $\Delta \bar{W}_i \in \mathbb{R}^{d \times d}$ with variable SR to the layers

Generate data $x \sim \mathcal{N}(0, I) \rightarrow \bar{f} \rightarrow y$. Train $f_0 \rightarrow f$ on (x, y) pairs and evaluate.



Conclusion

1. Lemma for optimal single-matrix approximation with $\text{rank} \leq R$ stable rank $\leq S$ proven
2. Previously known LoRA expressivity bounds for FNN and Transformer adjusted for constrained stable rank
3. LoRA generalisation gap upper bound tightened from $\mathcal{O}(\sqrt{R/N})$ to $\mathcal{O}(\sqrt{S/N})$
4. Stable rank empirically shown to be a better proxy for generalization and expressivity scaling